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***Helicobacter pylori* CagA Inhibits PAR1/MARK Family Kinases by Mimicking Host Substrates**

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Abstract

The CagA protein of *Helicobacter pylori* interacts with numerous cellular factors, and is associated with increased virulence and risk of gastric carcinoma. We present here the co-crystal structure of a subdomain of CagA with the human kinase PAR1b/MARK2, revealing that a CagA peptide mimics substrates of this kinase family, resembling eukaryotic protein kinase inhibitors. Mutagenesis of conserved residues central to this interaction renders CagA inactive as an inhibitor of MARK2.

Helicobacter pylori infects nearly 50% of the human population^{1,2} and has been closely linked to duodenal and gastric ulcers and adenocarcinomas¹. CagA is injected by *H. pylori* into the epithelial cells lining the stomach^{3–8}. Critical to many of the identified biological effects of the molecule on host cells, is the so-called “repeats domain,” a region with a strain-specific number of contiguous repeats of a 30–40 residue segment containing the EPIYA amino acid motif (Fig. 1a)⁷. The repeats domain interacts with and inhibits the PAR1/MARK (partitioning defective and MAP/microtubule affinity regulating kinases) family of protein serine/threonine kinases^{9–11}.

In order to understand the mechanism of CagA inhibition of PAR1/MARK kinases, we determined the 2.2 Å crystal structure of MARK2 in complex with a sub-domain of CagA spanning residues 885–1005 of Western *H. pylori* strain 26695, containing the A, B, and C EPIYA repeats (Fig. 1, a and b, Table 1, and Supplementary Methods). Surprisingly, the

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AUTHOR CONTRIBUTIONS D.N. performed all molecular biology, cloning, protein purification, functional assays, and crystallization, Z.T.Q. performed mass spectroscopy under the guidance of B.T.C., M.C.M. collected crystallographic data along with D.N., and C.E.S. solved the structure. All authors contributed to writing the manuscript.

The authors declare that they have no competing financial interests.

Accession codes. Protein Data Bank: Coordinates for the CagA-MARK2 complex have been deposited with accession code 3IEC.

majority of this 120 amino acid CagA domain was not visible in the crystals (although highly stable in complex with MARK2, and verified to be present by SDS-PAGE analysis of crystals, Supplementary Fig. 1a). In particular, the EPIYA motifs were disordered, and only a short 14 amino acid peptide possessed interpretable electron density (Fig. 1b and Supplementary Fig. 1b). The peptide does not adopt any clear secondary structure, but interacts with the kinase as an extended coil, burying approximately 950 Å² of surface area.

Two significant differences are present between the unbound¹² and the CagA-bound forms of MARK2, and both are structural hallmarks of kinases in their fully activated state. The first difference is an overall hinge motion between the N and C-terminal lobes of the kinase that brings them closer together in the presence of CagA (Supplementary Fig. 1c). This hinge motion is the same in each of the four independent copies of MARK2 in the asymmetric unit, making it unlikely that this is due to crystal packing. The second major difference is in the activation loop of the kinase, which adopts an ordered and activated structure in the presence of CagA (Fig. 1c)^{13,16}, including the conformation of the canonical Asp-Phe-Gly (DFG) motif that is required for magnesium binding, and the position of a threonine (Thr208 in MARK2) that is phosphorylated by activating kinases. These conformational states are remarkable for the fact that there is no nucleotide or magnesium present in the crystals, and no phosphorylation of Thr208.

What makes this activated conformation of the kinase possible, even in the absence of several elements normally required, is the CagA peptide. The visible peptide spans the sequence FPLKRHDKVDDLK, a repeat motif occurring twice in the crystallized construct. The peptide visible in the crystals, and which we show to be sufficient for inhibition of the kinase, we have termed “MKI,” for MARK2 Kinase Inhibitor, in analogy to PKI that inhibits PKA (*vide infra*).

Because the MKI sequence occurs twice in the crystallized construct, and only the amino acids common to both repeats are visible, it was unclear from the crystal structure which of the two possible peptide regions is binding (the first or second repeat of the MKI sequence). We addressed this issue through a combination of gel filtration chromatography and native mass spectrometry. These results (Supplementary Figs. 2-5, and Supplementary Tables 1-3) clearly demonstrate that each MKI sequence is bound by a molecule of MARK2.

The MKI peptide of CagA occupies the substrate-binding site of the kinase that is located near the interface between the N- and C-terminal lobes of the enzyme, using several amino acids to mimic conserved features of the PAR1 and AMPK family substrates (Fig. 2, a and b). The peptide is anchored to the kinase by four primary residues: Leu950, Arg952, Val956 and Leu959, numbering from the first repeat (Fig. 2a). Hydrophobic residues, especially leucine, are highly conserved at the corresponding positions in PAR1/MARK family kinase substrates (Fig. 2b), and the arginine at position 952 is also very well conserved. Several secondary interactions further stabilize the interaction - Phe948, His953, and Lys955, the last of which positions its terminal nitrogen atom in a location that mimics magnesium, forming hydrogen bonds with Asp193 of the MARK2 DFG motif (Fig. 2a). Overall, seven out of fourteen side-chains in the peptide interact with the kinase.

Intriguingly, the manner in which the CagA MKI sequence binds in the substrate-binding cleft is remarkably reminiscent of the manner in which PKI binds to and inhibits PKA (Fig. 2c, refs^{15,16}). A superposition of the two kinases bound to their inhibitors reveals that CagA residues 951-956 possess an overlapping main-chain conformation to residues 17-22 of PKI, and bind in a very similar location with respect to PKI in PKA (Fig. 2c). In addition to the location and main-chain conformational analogies, several side-chains of these kinase inhibitors interact with their targets in similar ways. For example, Arg18 of PKI is located very comparably to Arg952 of CagA (Fig. 2c), and both residues make hydrogen bonds with a conserved glutamic acid nearly identically positioned in the two kinases (Glu127 in PKA, and Glu136 in MARK2). Both peptides also use a short hydrophobic residue at the position of CagA Val956 (Ile22 in PKI) to insert into a conserved hydrophobic pocket in the kinases (Fig. 2c).

To test the importance of these side-chain interactions, a series of mutants were created in the MKI sequence of CagA. In order to prevent the second MKI sequence from biasing results, these mutants were made in a construct in which one MKI site was deleted (the construct spanning residues 885-981, see Fig. 1a), as well as in synthetic peptides corresponding to the minimal region defined by the crystal structure. Hexa-histidine-tagged CagA mutants were first examined for binding and co-elution with un-tagged MARK2 from Ni-NTA (Fig. 3a). **Point mutations of key anchoring residues, such as L950G and L959G, completely abolished binding to MARK2.** The R952G mutant exhibited weak binding (Fig. 3a), but interaction was highly unstable, however, and the complex was disrupted by ion exchange chromatography. The mutation V956G almost completely eradicated binding to MARK2, highlighting the importance of this hydrophobic interaction with the kinase. We also created two MARK2 mutants, encompassing CagA interacting residues E136G, F138G, and D139G in one construct (EFD), and L248G and D251G in the second (LD). EFD mutations completely abolished interaction between MARK2 and CagA, consistent with their interaction with key CagA binding residues (Leu950 and Arg952), whereas the LD mutants did not.

Both basal MARK2 kinase activity (Fig. 3b), as well as activated kinase activity using MARKK (Fig. 3c), were tested *in vitro* in the presence of varying concentrations of short peptides containing the wild type and mutant constructs of the MKI sequence. Synthetic peptides of CagA containing mutations in key interacting residues (Leu950, Arg952, Val956, or Leu959) failed to inhibit kinase activity except at extremely high concentrations (100µM). **In contrast, the wild type peptide and the K955G mutant were very efficient inhibitors of MARK2. Intriguingly, the K955G peptide was a slightly more potent inhibitor of MARK2 than wild type (Fig. 3c, d).** Supporting this data, East-Asian CagA subtypes contain glycine in the position that corresponds to Lys955 in Western CagA, and it has been reported that MARK2 binds more strongly to the East Asian CagA repeats region¹⁷.

This structure reveals that CagA mimics host substrates, using a short, 14 amino acid peptide (MKI) to bind to the kinase substrate-binding site (see also Supplementary Discussion and Supplementary Figure 6). Our biochemical experiments demonstrate that this peptide alone is sufficient to inhibit MARK2. In a dramatic example of convergent

evolution, *H. pylori* has evolved a peptide to mimic host substrates of this kinase family in order to manipulate eukaryotic cellular biochemistry during infection.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

Acknowledgments

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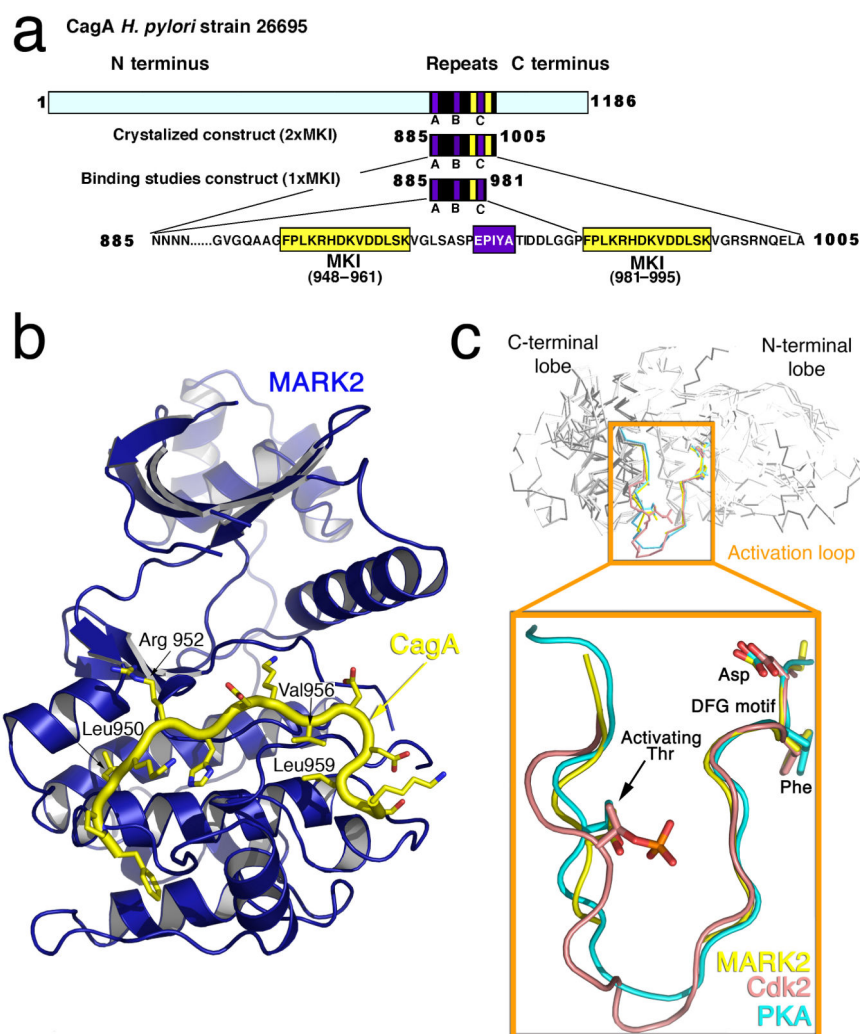


Fig. 1. Overall Structure of the CagA-MARK2 Complex. (a) Schematic representation of CagA. The A, B, and C EPIYA sequence repeats are shown as blue boxes. The crystallized construct (885-1005) and the deletion mutant used in binding studies that lacks one of the MKI sequences (885-981) are shown schematically as well. (b) Ribbon diagram of CagA-MARK2 complex, with MARK2 in blue, and the ordered MARK2 inhibitory sequence (MKI, MARK2 Kinase Inhibitor, residues 948-961 and 982-995), shown in yellow. (c) Alignment of several protein kinases, focusing on the activation loop. Cdk2 (PDB ID 1JST) and PKA (PDB ID 1ATP) are from structures of the kinases in activated states (including Cdk2 bound to cyclinA with activating phosphorylation of threonine).

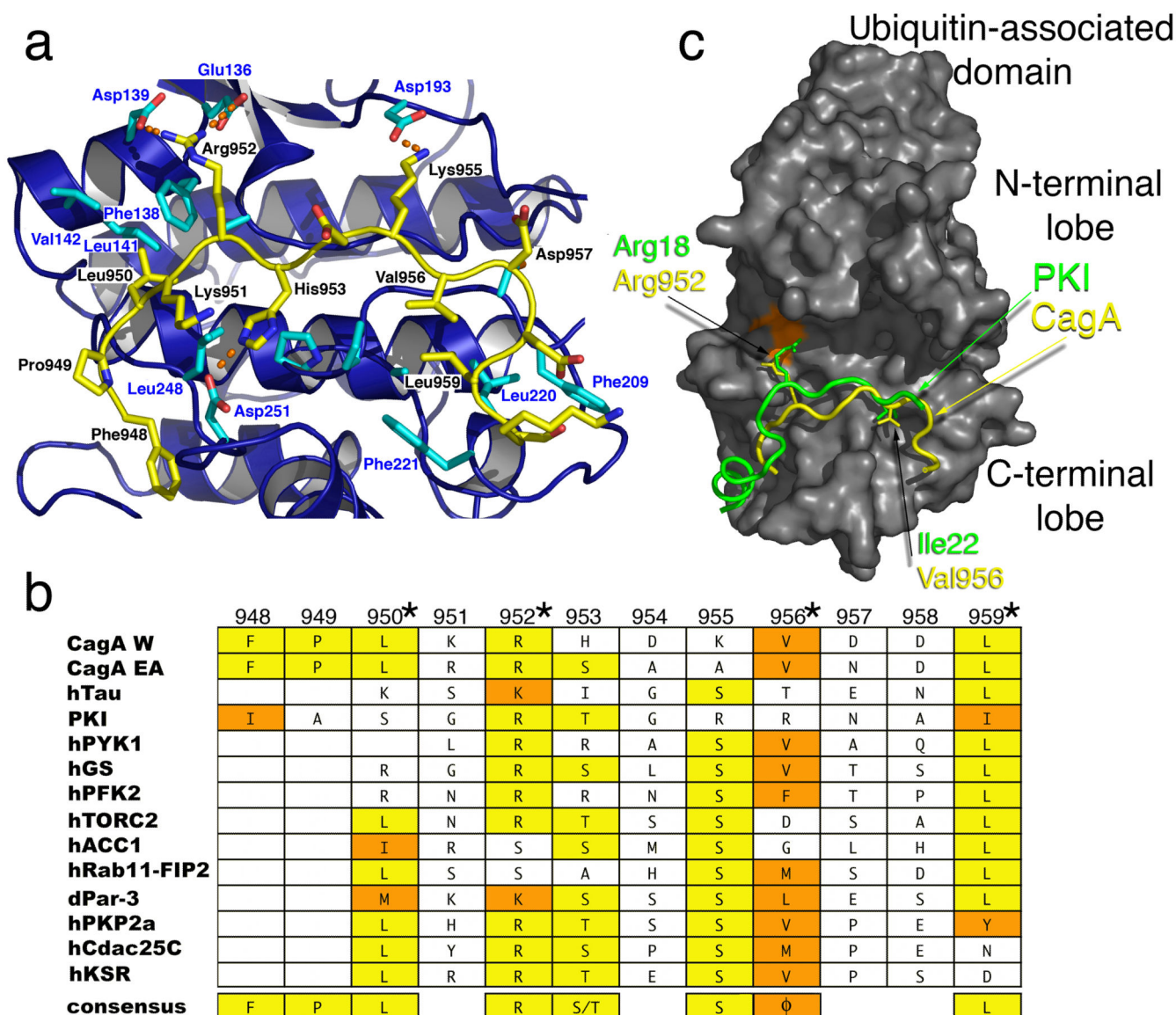


Fig. 2.

CagA is a Pathogenic Mimic of Host Substrates. (a) Details of the CagA peptide interaction. MARK2 in blue with cyan side-chains, while the MKI peptide of CagA in yellow. (b) Alignment of PAR1/MARK and AMPK family substrates with CagA peptide and, for comparison, PKI. Consensus identity is highlighted in yellow, and conservation in orange. Φ indicates a hydrophobic consensus. CagAW is CagA from *H. pylori* 26695 (Western subtype); CagA EA is Eastern-Asian subtype of CagA. (c) Superposition of PKI/CagA obtained from aligning the kinases PKA and MARK2. The surface of MARK2 is shown in dark grey. Glu136 of MARK2, which forms a salt bridge with CagA Arg952, is shown in orange on the surface of MARK2.

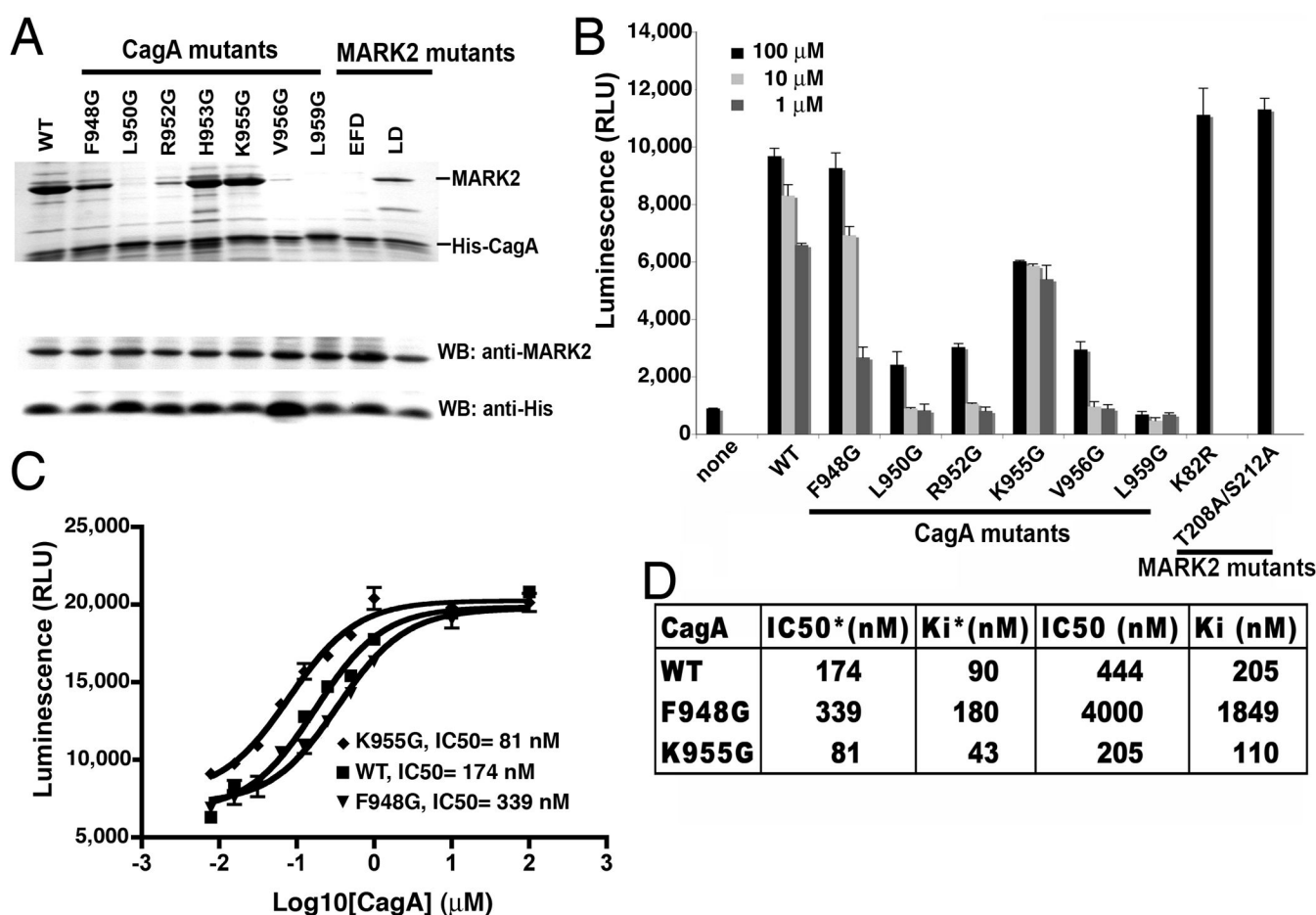


Fig. 3.

Mutational analysis of MKI mutants. (a) Binding of wild type or mutant hexahistidine-tagged CagA(885-981) to wild type or mutant MARK2(39-364) was assayed by pull-down experiments on Ni-NTA sepharose columns. Eluted material was subjected to SDS-PAGE and proteins stained with coomassie blue dye. Expression levels of MARK2 were determined by Western blot of total cell extract with anti-MARK2 antibody, and anti-His antibody for CagA expression. WT-wild type; EFD-MARK2 mutant (E136G, F138G, D139G); LD-MARK2 mutant (L248G, D251G). (b) Kinase activity of MARK2 in the presence of wild type or mutant CagA synthetic MKI peptides using a luminescent kinase assay. TR1 tau peptide (NVKSKIGSTENLK) at 500 μ M was used as a substrate with three different concentrations (100 μ M, 10 μ M, and 1 μ M) of CagA peptide inhibitors. MARK2 kinase-inactivated mutants, K82R and T208A/S212A, are negative controls. Error bars represent standard deviations from the mean. (c) Determination of IC₅₀ and K_i of CagA peptide inhibitors of MARK2 through a luminescence-based kinase assay. *-labeled values (IC₅₀* and K_i*) refer to kinase assays in which MARK2 was activated with MARKK. TR1 tau peptide was used as substrate (150 μ M- in the experiments with previously activated MARK2 or 200 μ M- in experiments without prior activation of MARK2), with increasing

concentration of inhibitory synthetic CagA peptides. Error bars represent standard deviations from the mean. (d) Table summarizing IC_{50} and K_i data.

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Table 1

Data collection and refinement statistics (molecular replacement)

Crystal 1 name	
Data collection	
Space group	$P2_1$
Cell dimensions	
a, b, c (Å)	93.47, 93.25, 113.95
α, β, γ (°)	90.00, 100.94, 90.00
Resolution (Å)	50-2.20 (2.28-2.20)
R_{sym} or R_{merge}	8.9 (70.6)
$I / \sigma I$	18.9 (2.62)
Completeness (%)	99.9 (100.0)
Redundancy	6.2 (6.2)
Refinement	
Resolution (Å)	41-2.20
No. reflections	97,422
$R_{\text{work}} / R_{\text{free}}$	20.4/24.8
No. atoms	
Protein	10,625
Water	687
B -factors	
Protein	25.82
Water	31.32
R.m.s. deviations	
Bond lengths (Å)	0.015
Bond angles (°)	1.507